

Experiment No. 5

ULTRASONIC INTERFEROMETER

Aim: To measure speed of ultrasonic waves in the given liquid.

Apparatus: Ultrasonic Interferometer, High frequency generator.

Theory: An ultrasonic interferometer is a device to measure speed of ultrasonic waves in liquids. Ultrasonic waves of known frequency are produced by quartz crystal fixed at the bottom of the cell. These waves are reflected by a movable metallic plate kept parallel to quartz crystal forming standing waves in the medium between the two. As the plate is moved up or down, consecutive nodes and antinodes are formed leading to corresponding minimum and maximum current in circuit. Hence, wavelength of the ultrasonic wave can be calculated.

Formula used: By knowing the wavelength of the ultrasonic waves, the velocity can be calculated using following relation:

$$v = \lambda f$$

Here, v = speed of ultrasonic waves (m/s)

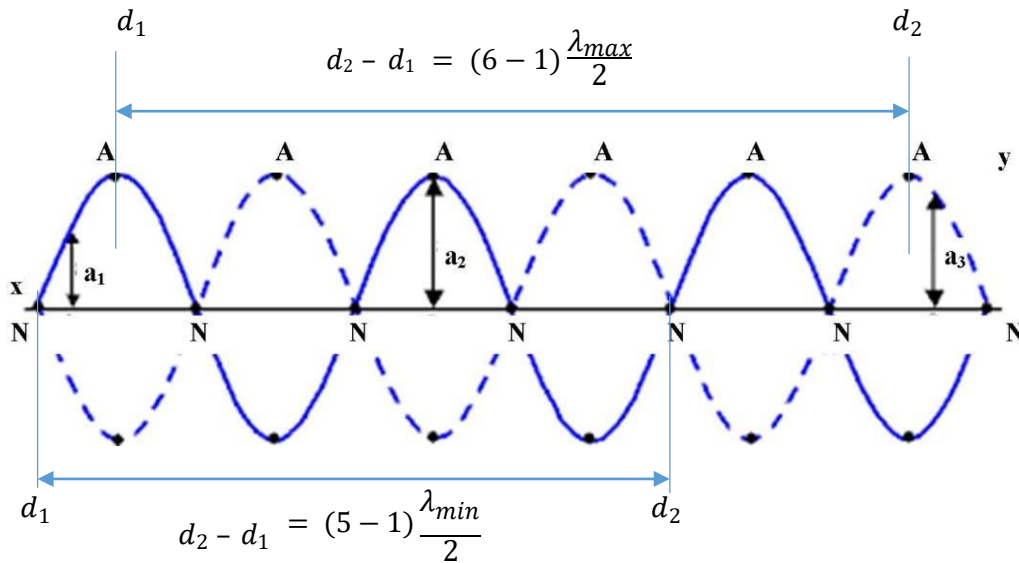
f = frequency of ultrasonic waves (Hz).

λ = Wavelength of ultrasonic waves (m).

$$\text{Least Count (L.C.)} = \frac{\text{Value of 1MSD}}{\text{No.of division on C.S.}}, \{ \text{Value of 1 MSD must have unit} \}$$

Procedure:

- 1) Make sure that cell at the bottom of interferometer is filled with distilled water and it is properly connected to the base socket.
- 2) Connect the high frequency generator with cell by co-axial cable provided with the instrument.
- 3) Move the micrometer slowly in clockwise direction till the anode current on the ammeter on the high frequency generator shows either maximum or minimum deflection
- 4) Note the readings of micrometer corresponding to the maximum or minimum in microammeter.
- 5) Measure the distance between 10-20 consecutive maxima or minima.
- 6) The distance (d) between ' n ' consecutive maxima or minima is $d = (n - 1) \frac{\lambda}{2}$
- 7) Find the average ($\lambda/2$) for the maxima and for minima. And hence calculate the wavelength by adding them.
- 8) Once the wavelength is known, speed of ultrasonic waves can be calculated using the given formula.



Observation:

For Maximum								
Sr. No.	Starting reading			(n) No. of maxima	Stop reading			$\left(\frac{\lambda}{2}\right)_{max}^d$ $= \frac{\lambda}{(n-1)}$
	MSR (mm)	VSR	TR (d ₁) (mm)		MSR (mm)	VSR	TR (d ₂) (mm)	
1								
2								
3								
4								
5								

Calculation:

Average $\left(\frac{\lambda}{2}\right)_{max} = \text{mm},$

Hence $\lambda = \left(\frac{\lambda}{2}\right)_{max} \times 2$

The speed of ultrasonic wave can be calculated using formula the formula $v = \lambda f$

Use $f = 2 \text{ MHz} = 2 \times 10^6 \text{ Hz}$

Error Analysis: Calculate the possible error due to measurement

Hint: Velocity is depending on wavelength and wavelength is depending on distance between nodes and antinodes. Relate the error with error in distances measured.

Note: convert the value of λ in m by multiplying the 2×10^{-3}

Result: Speed of the ultrasonic waves in given liquid is found to be _____ m/s

Precaution

1. Do not switch on the generator without filling the experimental liquid in the cell.
2. Remove experimental liquid out of the cell after use.
3. Keep it cleaned and dried.
4. Keep micrometer open at 25mm after use.
5. Avoid sudden rise or fall in the temperature of circulated liquid to prevent thermal shock to the quartz crystal.
6. While cleaning the cell care should be taken not to spoil or scratch the gold plating on the quartz crystal.
7. Give your generator 15 seconds warming up time before the observation.